
CS 421 --- State Activity

Name	Netid

Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

Challenge

Example 1) Can you write a counter function (or function producer) without using objects and without using global variables?

Two hints: python supports HOFs and nested function declarations. To access an outer scoped variable, use the `nonlocal` keyword. Here is a convoluted ``increment'' function that shows some of the features you will need.

```
0 def inc(x):
1     i = x
2     def doit():
3         nonlocal i
4         i = i + 1
5
6     doit()
7     return i
```

Problem 1) Consider this Python code.

```
0 class Delay:
1     def __init__(self, action):
2         self.action = action
3         self.status = 0
4
5     def report(x):
6         print("Thunk executed: {}".format(x))
7         return x
8
9     def force(self):
10        if self.status == 2:
11            return self.value
12        elif self.status == 0:
13            self.status = 1
14            self.value = Delay.report(self.action())
15            self.status = 2
16            return self.value
17        else:
18            return Exception("It broke!")
```

- Review this code with a partner. Can you figure out how to use it?
- What is the purpose of `self.status = 1`?

Consider this function and list definition.

```
0 def lazyTake(n,x):  
1     if x==() or n<1:  
2         return ()  
3     else:  
4         return (x[0],lazyTake(n-1,x[1].force()))  
5  
6 l1 = (2,Delay(lambda: (3, Delay(lambda: (5, Delay(lambda: ()))))))
```

Problem 2) Can you write lazyTail, lazyMap, and lazyZipWith?

Problem 3) Use these functions to make the infinite list of natural numbers and Fibonacci numbers.